Fateme Mazdarani

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Research Interests

My research broadly focuses on artificial intelligence and machine learning, particularly the reliability, robustness, and interpretability of intelligent systems. I study large language models and mathematical reasoning, including how models access, evaluate, and verify knowledge across different representations and how retrieval can support more reliable reasoning. My broader interests include human-centered and trustworthy AI, recommender systems, data mining, and scalable machine learning methods for real-world applications.

Education

Clemson University, Ph.D. in Computer Science Jan 2025 – Present

- Advisor: Dr. Carlos Toxtli, Human-AI Empowerment Lab
- Research: robust and trustworthy LLM evaluation, mathematical reasoning, and human-centered AI
- GPA: 4.0/4.0

University of Tehran, B.Sc. in Computer Engineering Aug 2018 – Jun 2023

- GPA: 3.5/4.0
- Bachelor's thesis: A Multi-Platform Voice Recognition System Activator in Persian Language

Publications and Manuscripts

Published

- F. Mazdarani, A. C. Hernandez, and C. Toxtli. **Worker-Centered AI: Transparent Explanations for Trustworthy Task Recommendation in Crowd Work.** *2026 IEEE Conference on Artificial Intelligence (CAI)*, pp. 295–302, 2026. DOI: 10.1109/CAI68641.2026.11536153.

Accepted for Publication

- F. Mazdarani and C. Toxtli. **Beyond the Answer Key: Robustness Evaluation of Large Language Models for Step-Level Mathematical Verification.** *2026 IEEE International Conference on Machine Learning and Applications (ICMLA).* Accepted for publication.
- F. Mazdarani and C. Toxtli. **Randomized SVD Approximations for Spectral Co-Clustering of Word-Document Matrices.** *2026 IEEE International Conference on Machine Learning and Applications (ICMLA).* Accepted for publication.
- F. Mazdarani and C. Toxtli. **TREAT: Evaluating Access to Formal Knowledge across Equivalent Mathematical Representations.** *2026 International Symposium on Symbolic and Numeric Algorithms for Scientific Computing (SYNASC).* Accepted for publication.
- F. Mazdarani and C. Toxtli. **The Shape of Mathematical Creativity: Measuring Mathematical Exploration in Formal Proof Generation.** *MathAI Workshop at NeurIPS 2026.* Accepted for publication.
- F. Mazdarani and C. Toxtli. **From Recognition to Reconstruction: Towards Robust Procedural Mathematical Reasoning.** *The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026); Workshop on Mathematical Natural Language Processing.* Accepted for publication.

Under Review

- F. Mazdarani and C. Toxtli. **The Models Behind the Claim: Empirical Standards for LLM Evaluation.** *International Conference on Learning Representations (ICLR).* Under review.

Honors and Awards

- Dr. Robert M. Geist III Endowed Fellowship in Computing, Clemson University, 2025–2026
- Ranked 79th in the National University Entrance Exam in Iran among more than 300,000 participants, 2018
- Accepted in the First Round of the Iranian National Mathematics Olympiad, 2015–2016

Research Experience

Graduate Research Assistant Fall 2026 – Present
Human-AI Empowerment Lab, Clemson University
Research in the lab since August 2025.

- **LLM reasoning and evaluation:** develop benchmarks for theorem recognition across equivalent representations, step-level verification, and retrieval-supported procedural reasoning.
- **Autonomous underwater vehicles (AUVs):** develop and evaluate vision-language-action navigation in simulation, integrating object detection and safety checks; compare grid-based and learned controllers for search coverage, target acquisition, travel efficiency, and inference latency.
- **Worker-centered AI:** study how explanations and navigation cues affect fairness, trust, empowerment, usability, and task selection in crowd work.
- **Scalable machine learning:** investigate randomized SVD for spectral co-clustering of word-document matrices and ADMM-based sparse spectral clustering.

Teaching Experience

Graduate Teaching Assistant, Clemson University 2025 – 2026

- Served as teaching assistant for Design and Analysis of Algorithms and Human-Computer Interaction / Human-Centered Computing courses.
- Supported grading, rubric-based feedback, project checkpoints, student questions, and course assessment.

Teaching Assistant, Engineering Probability and Statistics, University of Tehran Fall 2021
Teaching Assistant, Discrete Mathematics, University of Tehran Fall 2019 – Spring 2021

Academic Service

Social Chair Summer 2026 – Present
School of Computing Graduate Student Association, Clemson University
Delegate Fall 2026 – Present
Clemson Graduate Student Government (GSG)
Peer Reviewer
Reviewed 10+ papers at various conferences.
Poster Judge August 2026
Creative Inquiry undergraduate research program, Clemson University

Selected Presentations

- Poster presentation: "Beyond the Answer Key: Robustness Evaluation of Large Language Models for Step-Level Mathematical Verification," *2026 IEEE International Conference on Machine Learning and Applications (ICMLA)*, 2026.
- Poster presentation: "Randomized SVD Approximations for Spectral Co-Clustering of Word-Document Matrices," *2026 IEEE International Conference on Machine Learning and Applications (ICMLA)*, 2026.
- Conference paper presentation: "TREAT: Evaluating Access to Formal Knowledge across Equivalent Mathematical Representations," *2026 International Symposium on Symbolic and Numeric Algorithms for Scientific Computing (SYNASC)*, 2026.
- Virtual paper presentation: "Worker-Centered AI: Transparent Explanations for Trustworthy Task Recommendation in Crowd Work," *2026 IEEE Conference on Artificial Intelligence (CAI)*, online, 2026.
- Poster presentation: "Worker-Centered AI: Transparent Explanations for Trustworthy Task Recommendation in Crowd Work," *2026 INTEGRATE Symposium*, University of Wisconsin–Madison, 2026.

Skills

- **Programming Languages:** Python, MATLAB, Java, C/C++, R, SQL, Assembly (Intel x86, MIPS)
- **Machine Learning and Data Science:** PyTorch, TensorFlow, Keras, NumPy, pandas, scikit-learn, Matplotlib, statistical analysis, experiment design
- **LLM and AI Evaluation:** prompt engineering, LLM-as-a-judge evaluation, process-level evaluation, benchmark design, Hugging Face Transformers, OpenAI API, Gemini, Claude
- **Web, Scripting, and Tools:** HTML, CSS, JavaScript, React, LaTeX, Bash, Git, Docker, Maven
- **Hardware and Simulation Tools:** Arduino, ModelSim, Intel Quartus Prime, LTspice, Proteus circuit simulator

Additional Research Projects and Coursework

Quantum Machine Learning for Cybersecurity 2025

- Implemented and evaluated hybrid quantum deep learning and variational quantum classifier-based models for botnet DGA attack detection.

Sparse Spectral Clustering 2025

- Studied ADMM-based optimization methods for sparse spectral clustering.

Persian-Language Voice Activator 2022 – 2023

- Trained an RNN using Keras and TensorFlow for a multi-platform voice recognition system activator in Persian; bachelor's thesis, University of Tehran.

Movie Information Management System 2022

- Implemented a full-stack course project using Java, React, REST APIs, SQL, Maven, Git, unit testing, and Tomcat.

CNN Acceleration by Removing Zero-Valued Neurons 2022

- Wrote a research paper for the Research and Technical Presentation course.

Smart Irrigation System 2022

- Simulated sensors and electric motors using Arduino IDE and Proteus for a real-time embedded systems project.

Android Surface Scanner 2022

- Developed an Android-based surface scanner using Java and Android Studio.

SIMD Instructions for IP Acceleration 2021

- Implemented acceleration techniques using C++ and Intel SIMD instructions.